

Lab 1

Railway ticketing portal

The system portal includes distance-based pricing for train tickets. The system offers basic and advanced search capabilities. It allows users to search trains according to various options, including a one-way or round trip and multi-city.

The system also has dynamic pricing rules that include conditions such as specific discounts. If you take the train before 9:30 am or in the afternoon after 4:00 pm until 7:30 pm ('the rush hour'), you must pay the full fare. A saver ticket is available for trains between 9:30 am and 4:00 pm, and after 7:30 pm, you get a 5% discount. If you hold an 'over 60s' rail card, you get a 34% discount on whatever ticket you buy. If you are traveling with a child (under 16), you can get a 50% discount on any ticket if you hold a family rail card, otherwise you get a 10% discount. You can only hold one type of rail card. The tickets can be added to a shopping cart in a single booking, which users have access to for a whole week.

The system allows users to see previous reservations across various booking channels, and to create, modify and cancel them.

The system offers customer profile management as users can add, edit, and view consumer profiles.

Various passenger types can be managed, with their relevant concession, and promotions and discount coupons being applied during the booking process.

The multi-language feature allows customers to select a required language. Data will be displayed in languages such as Spanish, German, French, and English, and additional languages can be requested.

Automated supplier payments, reconciliation and adjustments are also available and can be integrated with external accounting systems, including SAP, Intuit QuickBooks and Microsoft AX.

1. Create a detailed list of functional and non-functional software requirements for the Railway ticketing portal system.

2. Write 3 *Features* using Gherkin Syntax.